Data Analyst Assessment

BrightLearn (Pty) Ltd

Thank you for applying. Answer the questions in order, and skip any you can't answer.

We'd like to see your own work on this one, so please don't use AI assistants. We use them here day to day, and this exercise only tells us something if it's yours. You own every number you submit, and we'll walk through the work with you afterwards.

Setup

1. Install DuckDB

DuckDB version 1.1 or later.

- macOS: `brew install duckdb`
- Windows: `winget install DuckDB.cli`
- Any platform: duckdb.org/install, or `pip install duckdb`

2. Build the database

Open a terminal in the unzipped folder and run:

```
duckdb assessment.duckdb < load_data.sql
```

On Windows PowerShell:

```
duckdb assessment.duckdb -c ".read load_data.sql"
```

Check it:

```
SELECT COUNT(*) FROM fact_sales_line;
```

You should get 315,182 rows.

All tables are in the default schema. Create your views there and keep their definitions in `views.sql` as you go.

Setup problems aren't part of the assessment. If something won't work, email us and we'll sort it out or send the data another way.

The data

The data is a retail pharmacy group with 40 stores across South Africa, selling front shop, dispensary, beauty and baby ranges, with a loyalty programme called the Wellness Club. Every row is synthetic.

The first eight tables are its gold layer. The last two are not. Those two are what the business works from today, passed on as they were received. Nobody wrote down where either came from.

Table	Grain	Notes
`fact_sales_line`	One product line on a transaction	`customer_key` is empty for walk-in trade
`fact_campaign_audience`	One member selected for a campaign	`audience_group` records how the member was treated
`dim_customer`	One member per tier version	`tier_valid_from` and `tier_valid_to` bound each version. `email_hash` hashes the member's email
`dim_product`	One product	`is_script` marks prescription medicine
`dim_store`	One store	
`dim_date`	One calendar day	Calendar month and the 4-4-5 fiscal calendar
`legacy_reports`	One report in the reporting estate	Definitions as their owners wrote them, with 90-day usage
`legacy_report_figures`	One figure a legacy report last published	
`product_cost_greg`	One cost price per product, as received	Kept by hand in Merchandise. The only cost figures in the data
`mkt_category_month`	One category per calendar month	Marketing's own sales summary

Amounts are in rand. `net_amount` is `gross_amount` less `discount_amount` plus `dispensing_fee`.

The front shop sets its own prices and runs promotions. Prescription medicine is a regulated fee business. On a product with `is_script`, `list_price` is its Single Exit Price, `fee_band` is the band that price falls in, and `dispensing_fee` is the fee the pharmacy is paid for filling the script. `discount_funder` records who paid for a discount.

The Wellness Club reviews tiers on 1 January and 1 July against the member's net spend over the previous 26 weeks. A review that changes a tier closes the row in `dim_customer` and opens a new one.

The fiscal year is a 4-4-5 retail calendar. It starts on the Monday of the week holding 1 March, is named for the calendar year in which it ends, and its quarters run four, four and five weeks.

The questions

Each answer needs a SQL view in your DuckDB file that produces the numbers you report. Write SQL that would also run on Databricks SQL, so please avoid DuckDB-only shortcuts such as `SUMMARIZE`, `PIVOT`, `EXCLUDE` and list comprehensions.

1. The Head of Retail asks for net sales by category for each fiscal period of FY2026 and for the FY2027 periods so far.
2. The Wellness Club manager asks how many members we have and what a member spends with us in a year.
3. The Merchandise director asks where net sales growth has come from over the 18 months of history, by category and by store, and whether the growth came from price or from volume.
4. The Head of Retail asks which categories earn us the most, and says Marketing's summary already answers it. Give them your answer and say how much of it you would defend. Account for every rand of the difference between their summary and the gold layer, and say which of the two you would report. Where the data cannot answer the question, write down your assumption and who you would ask about it.
5. Marketing mailed a Wellness Club activation to members in March 2026. They say the members who received it spent over 180% more than members who did not, and they want to repeat it at a larger scale and budget. Tell them what the campaign was worth and whether the evidence supports scaling it up. Then work out the holdout the next campaign needs to detect a 5% uplift.
6. The analytics chapter lead asks you to look at the twelve reports in `legacy_reports`, name the three you would retire first, and name the ones whose definitions conflict with the gold layer. Then take R01. Its owner says the March 2026 figure is right and the gold number for the fiscal period is wrong. Explain the difference to them line by line, in business terms.

End with two or three recommendations that your evidence supports.

What to send back

1. Your views in `views.sql`, runnable top to bottom on a fresh load.
2. A `SUBMISSION.md` that says how to run your work and names the recurring metrics you would ask an Analytics Engineer to certify, each with its grain and its definition. Write your assumptions and judgment calls in it as well, and anything you found that someone else needs to fix. We read that part as closely as the SQL.
3. A dashboard with the answers you would put in front of the Head of Retail every month. Build it in any free tool. Metabase, Apache Superset and Rill all read a DuckDB file directly.
4. A three-minute screen recording of you using that dashboard, with the link in `SUBMISSION.md`. Change something in it while you record.

A rough recording is fine. Loom's free plan runs to five minutes and whoever watches doesn't need an account, and any other recorder works. Share the one window you are working in.

Send the rest as one zip or a link to a private repository, since the upload rejects loose script files. To make the zip, put your files in one folder, then right-click it and choose Compress on macOS, or Send to and Compressed (zipped) folder on Windows.